



Canvas Course Manager

User Guide

Version 2.0



Harvard University

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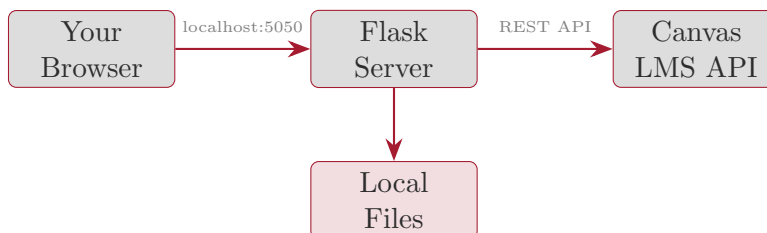
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Getting Started

Overview

Canvas Course Manager is a locally-run web application that gives instructors a unified interface for managing Canvas LMS courses. Instead of navigating between multiple Canvas web pages, you get a single dashboard with tabs for browsing course content, managing files, editing assignments and quizzes, tracking student progress, and viewing a course calendar.



The Flask server acts as a proxy between your browser and the Canvas API, handling authentication and parallel data fetching to keep the UI responsive.

Installation

1. Clone the repository:

```
git clone https://github.com/astrostubbs/CanvasFrontEnd.git
cd CanvasFrontEnd
```

2. Install Python dependencies:

```
pip install -r requirements.txt
```

3. Configure your Canvas API token (see Section 1.3).

4. Run the server:

```
set -a; source .env; set +a
python canvas_gui.py
```

5. Open <http://localhost:5050> in your browser.

Obtaining a Canvas API Token

1. Log in to your Canvas instance (e.g., canvas.instructure.com).
2. Navigate to **Account** → **Settings**.
3. Scroll down to **Approved Integrations**.

4. Click **+ New Access Token**.
5. Enter a purpose (e.g., “Course Manager”) and optionally set an expiry date.
6. Click **Generate Token** and copy the token immediately.

Important

Your API token grants full access to your Canvas account. Never commit it to version control or share it. Store it in a `.env` file (which is git-ignored) or export it as an environment variable.

Environment Configuration

Copy the example configuration and fill in your values:

```
cp .env.example .env
```

Edit `.env` with your settings:

```
CANVAS_API_TOKEN=1234~AbCdEfGhIjKlMnOpQrStUvWxYz
CANVAS_API_URL=https://canvas.instructure.com/api/v1
LOCAL_ROOT=~/Documents/MyCourse
```

Table 1: Environment Variables

Variable	Required	Description
CANVAS_API_TOKEN	Yes	Your Canvas personal access token
CANVAS_API_URL	Yes	Canvas API base URL (with <code>/api/v1</code>)
LOCAL_ROOT	No	Root directory for local file browser

The Interface

When you first open the application, you see a header with a course selector and a row of seven tabs. Select a course from the dropdown to populate all tabs with that course’s data.



The tabs are:

1. **Browse** — Course structure overview with expandable modules
2. **Calendar** — Month-by-month calendar of assignments and quizzes
3. **Students** — Student performance analytics and progress tracking
4. **Files** — Two-panel file manager (local + Canvas)
5. **Assignments** — Full-screen assignment editor
6. **Quizzes** — Full-screen quiz and question editor
7. **Page Editor** — Create and edit Canvas pages
8. **LaTeX → Canvas** — Convert L^AT_EX equations to Canvas format with live preview
9. **AI Assistant** — Chat interface powered by Claude

Browse Tab

The Browse tab shows a hierarchical view of your course content.

Global Links

At the top, quick-access links open the Canvas web interface for common pages: the course home, gradebook, syllabus, all files, and the people list.

Module Tree

Below the links, each module appears as an expandable row. Click a module to load and display its contents, **grouped by type**:

The screenshot shows a module titled "Module 2: Data Analysis" with a red header bar and a "8 items" badge. Below the header, the contents are grouped by type:

- ASSIGNMENTS (3)**
 - Problem Set 2: Regression
 - Lab: Linear Models
 - Homework 2 Revisions
- QUIZZES (1)**
 - Module 2 Review Quiz
- PAGES (2)**
 - Lecture Notes: OLS
 - Reading: Chapter 4
- FILES (2)**
 - dataset.csv, notebook.ipynb

Tip

Click any item to open it directly in Canvas in a new browser tab. Items are lazy-loaded — the module contents are only fetched from Canvas when you click to expand the module.

Calendar Tab

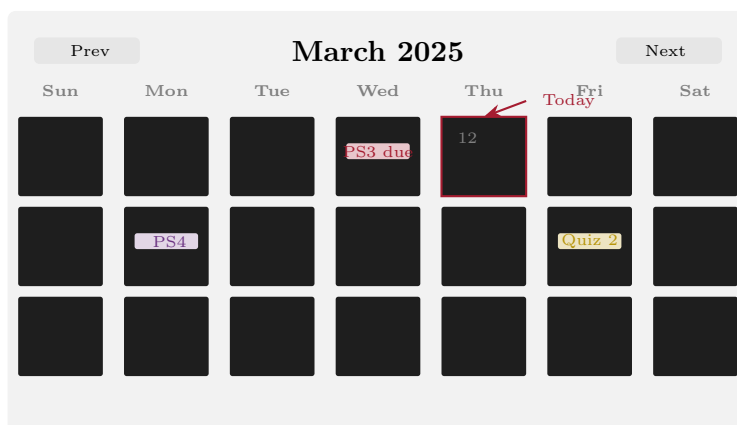
The Calendar tab provides a month-by-month view of your course schedule.

Navigation

Use the **Prev/Next** buttons to move between months, or click **Today** to jump to the current month.

Events

- **Assignment due dates** appear as red-tinted badges on their due date.
- **Availability windows** are shown as purple markers on the unlock date (when the assignment becomes available).
- **Quiz due dates** appear as amber/yellow badges.



Tip

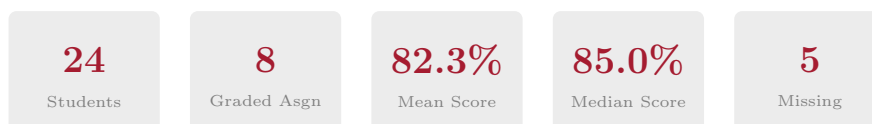
Click any event badge to jump directly to that item's editor within the application (not Canvas). For example, clicking a quiz due date on the calendar switches to the Quizzes tab and opens that quiz. Hover over a badge to see the full name and point value.

Student Progress Tab

The Students tab provides a comprehensive analytics dashboard for tracking student performance. This is designed to answer the questions instructors ask most often: Who is falling behind? What's the grade distribution? Which assignments are students struggling with?

Summary Statistics

Five summary cards appear at the top:



Ranked Student Scores

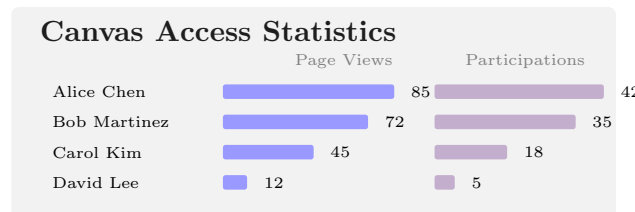
A horizontal bar chart shows every student's current percentage score, ranked from highest to lowest. Bars are color-coded:

- **Green** ($\geq 90\%$),
- **Yellow** (70–89%),
- **Red** ($< 70\%$).

This makes it easy to spot students who may need additional support.

Canvas Access Statistics

A dual bar chart showing each student's **page views** (blue) and **participations** (purple), sorted by total activity. This gives you an at-a-glance view of who is actively engaging with the course material and who may need outreach.



Tip

Low page views combined with low scores may indicate a student who isn't engaging with the material. High page views with low scores may indicate a student who is struggling despite effort.

Missing Assignments

A sorted list of students who have missing submissions, ordered by the number of missing assignments (most first). Hover over a row to see which specific assignments are missing.

Important

The "missing" flag comes from Canvas. An assignment is marked missing if it is past due and the student has not submitted anything. If you use "on paper" submission types, Canvas may incorrectly flag those as missing.

Late Submissions

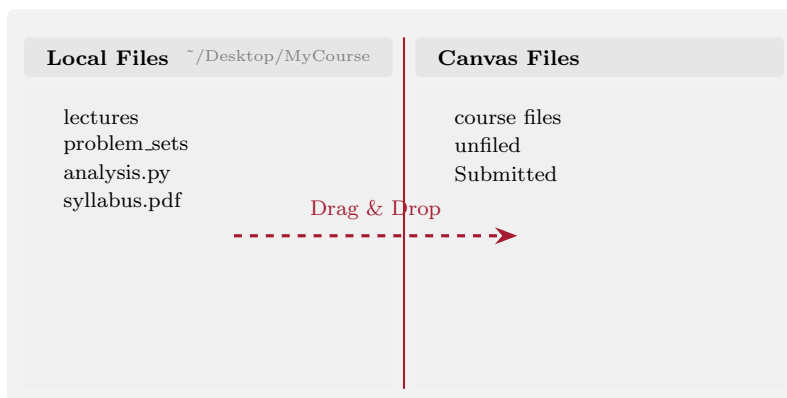
Similar to the missing list, this shows students with late submissions and the count of late work.

Per-Assignment Averages

A bar chart showing the class average score for each graded assignment. This helps identify assignments that were particularly challenging (low average) or straightforward (high average).

File Transfer

The Files tab provides a two-panel file manager: your local files on the left, Canvas course files on the right.



Uploading Files to Canvas

1. In the left panel, locate the file you want to upload.
2. **Drag** the file from the left panel and **drop** it onto a Canvas folder in the right panel.
3. A confirmation dialog appears letting you choose the destination folder.
4. Click **Upload** to transfer the file.

The upload uses Canvas's three-step upload process (notify, upload, confirm) and handles it automatically.

Browsing Canvas Files

Click any folder in the right panel to expand it and see its contents. Click a file to open/-download it in a new tab.

Local File Navigation

The local file panel includes navigation controls at the top:

- **↑ Up** — go to the parent directory.
- **Home** — go to your home directory (~).
- **Desktop** — go to ~/Desktop.
- **Click path display** — type any path to navigate directly.
- **Double-click a folder** — enter that folder as the new root.

File Type Filtering

Both panels include a filter bar with buttons: **All**, **PDF**, **Python**, **Images**, **Docs**, **Data**, **Code**. Click a filter to show only matching file types. For example, clicking **PDF** hides everything except .pdf files.

Resizable Panels

The vertical divider between the two panels (and between sidebars and editors in other tabs) is draggable. Grab and drag it to resize the panels to your preference.

Tip

The local file panel starts at the directory specified by the `LOCAL_ROOT` environment variable. Set this to your course's working directory for quick access. You can navigate away at any time using the navigation buttons.

Assignment Editing

The Assignments tab provides a full-screen editor with a sidebar list and a main editing area.

The screenshot shows the 'Edit Assignment' interface. On the left is a sidebar with a list of assignments: 'Problem Set 1', 'Problem Set 2' (highlighted), 'Lab Report', and 'Final Project'. At the top of the sidebar is a '+ New' button. The main area is titled 'Edit Assignment' and contains a form for 'Problem Set 2'. The form includes fields for 'DUE DATE' (Mar 15), 'AVAILABLE FROM' (Mar 1), and 'UNTIL' (Mar 20). At the bottom of the main area are three buttons: 'Save' (green), 'Unpublish' (grey), and 'Delete' (red).

Editing an Existing Assignment

1. Click an assignment in the sidebar to load it into the editor.
2. Edit any field: name, description, due date, availability window, points, grading type.
3. A green **Save Changes** button appears when you modify any field.
4. Click **Save Changes** to push the update to Canvas.

Creating a New Assignment

1. Click the + **New** button at the top of the sidebar.
2. Fill in the assignment name (required), description, due date, points, submission type, and grading type.
3. Click **Create Assignment**. It will be created as a draft (unpublished).
4. Publish it when ready using the **Publish** button in the editor.

Split HTML Editor

The assignment description uses a side-by-side editor:

- **Left:** Raw HTML source (editable textarea).
- **Right:** Live rendered preview that updates as you type.

This means you can write HTML tags like `<p>`, `<ol>`, `<strong>` in the source panel and immediately see how they'll appear to students.

Example: Assignment Description HTML

```
<p>This problem set covers <strong>orbital mechanics</strong>.</p>
<ol>
  <li>Derive Kepler's third law from Newton's law.</li>
  <li>Calculate the escape velocity from Earth.</li>
  <li>Plot the trajectory of a satellite using Python.</li>
</ol>
<p><em>Submit as a single PDF.</em></p>
```

AI-Generated Assignments

Click the **AI Generate** button in the sidebar header to open the AI assignment generator:

1. Enter a **topic or instructions** describing what the assignment should cover.
2. Optionally add **source materials** (Canvas pages, local files, or free text) to give the AI context.
3. Select the **assignment type** (Problem Set, Essay, Lab Report, Project).
4. Set the **points possible**.
5. Click **Generate Assignment**.
6. Review and edit the generated title, description (in the split editor), and points.
7. Click **Create Draft Assignment in Canvas** to create it as an unpublished draft.

The screenshot shows the 'AI Assignment Generator' interface. At the top, it says 'AI Assignment Generator' in red. Below that is a section labeled 'TOPIC / INSTRUCTIONS' with a text input field containing 'Create a problem set on stellar evolution, HR diagram, main sequence lifetimes'. Underneath is a section labeled 'SOURCE MATERIALS' with two items: 'Lecture 5 Notes' and 'chapter4.pdf', each with a close button (x). Below the source materials are two input fields: 'Problem Set' and '100 pts'. At the bottom is a large red button labeled 'Generate'.

AI Review of Assignments

When editing an existing assignment, click the **AI Review** button in the header. The AI will analyze the assignment description and identify:

- Ambiguous or unclear instructions
- Missing information students would need
- Unclear grading criteria
- Wording that could be improved

The review streams in below the assignment editor. Use the feedback to refine your assignment before publishing.

Publishing and Deleting

- **Publish/Unpublish:** Toggle the publication state directly from the editor.
- **Delete:** Click the red **Delete** button. A confirmation dialog will appear.
- **Open in Canvas:** A link button opens the assignment in the Canvas web UI.

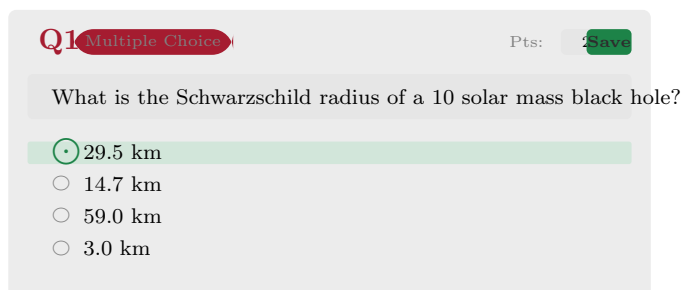
Quiz Editing

The Quizzes tab has the same sidebar-plus-editor layout. Selecting a quiz loads all its questions as editable cards.

Editing Questions

Each question card shows:

- The question number and type (Multiple Choice, True/False, Essay, etc.)
- An editable text area for the question text (supports HTML)
- An inline points input
- For multiple-choice questions: radio buttons to select the correct answer, with editable answer text fields



The screenshot shows a quiz question card with a light gray background. At the top left, it says "Q1 Multiple Choice" in red. At the top right, it says "Pts: 2" followed by a green "Save" button. The question text is "What is the Schwarzschild radius of a 10 solar mass black hole?". Below the question, there are four radio button options: "29.5 km" (which is selected and highlighted with a green bar), "14.7 km", "59.0 km", and "3.0 km".

Workflow: Edit and Save

1. Modify the question text, points, or answer text.
2. For multiple-choice: click the radio button next to the new correct answer.
3. The card border turns **red** and a **Save** button appears.
4. Click **Save** to update the question on Canvas.

Note

Published quizzes *can* be edited via the API as long as no student has started a submission attempt. If students have already taken the quiz, Canvas may restrict certain changes.

Adding Questions (Hierarchical)

Click **+ Add Question** to open a two-step form:

Step 1: Choose the question type from a visual grid:

Multiple Choice 4 options, 1 correct	True / False Two options	Short Answer Student types text
Essay Free-form writing	Numerical Student enters a number	Multiple Answers Select all that apply

Step 2: Fill in the type-specific form. The form adapts to the question type:

- **Multiple Choice:** Radio buttons for correct answer selection, 4+ editable answer fields.
- **True/False:** Pre-filled True and False options; select which is correct.
- **Short Answer / Numerical:** Text fields for acceptable answer(s) (all marked correct).
- **Essay:** No answer fields — just the question text and points.
- **Multiple Answers:** Checkboxes (select *all* correct answers).

The question text field includes a **side-by-side HTML preview**, so you can use `<p>`, `<strong>`, equation image tags, etc. A “Change Type” link lets you go back to Step 1 without losing your text.

Creating a New Quiz

Click **+ New** in the sidebar to open the quiz creation form. Set the title, description, quiz type (graded, practice, survey), and optional time limit. New quizzes are created as drafts — publish when ready.

AI-Generated Quizzes

Click the **AI Generate** button in the sidebar header to open the quiz generator:

1. Enter a **topic or instructions** (e.g., “Stellar evolution, HR diagram, main sequence lifetimes”).
2. Optionally add **source materials** by clicking **+ Add Sources**:
 - **Canvas Pages** — select from a checklist of your course pages.
 - **Assignments** — select from a checklist of assignment descriptions.
 - **Local Files** — browse your filesystem and check files to include.
 - **Free Text** — paste chapter notes, topic lists, or custom context.
3. Choose a **prompt template** (default: “Generate Quiz Questions”).
4. Set the **number of questions**, **question type** (MC, T/F, Short Answer, Essay, or Mix), and **points per question**.
5. Click **Generate Questions**.
6. Review the generated questions in editable cards (with side-by-side HTML preview).
7. Edit question text, change correct answers, adjust points, or delete individual questions.
8. Click **Create Draft Quiz in Canvas** to create an unpublished quiz with all questions.

The screenshot shows the 'AI Quiz Generator' interface. At the top, the title 'AI Quiz Generator' is in red. Below it, the 'TOPIC' field contains the text 'Stellar evolution: HR diagram, main sequence lifetimes, post-MS evolution'. Under 'SOURCE MATERIALS', there are two items: 'Lecture 7 Notes' and 'Ch5_stellar.pdf', followed by a '+ Add Sources' button. Below these fields, there are four buttons: 'Generate Quiz Questions', '5 questions', 'Multiple Choice', and '1 pt each'. A large red 'Generate' button is positioned below the configuration buttons. At the bottom, a dashed box contains the text '↓ Generated questions appear here as editable cards'.

Tip

The AI quiz generator uses prompt templates that you can customize. Click the gear icon in the header to open the Prompt Editor. The “Generate from Reading” template works best when you add source materials.

AI Review of Quizzes

After loading a quiz's questions, click the **AI Review** button in the quiz header. The AI examines every question and its answers, identifying:

- Ambiguous or unclear wording that could confuse students
- Questions where multiple answers could reasonably be considered correct
- Trick questions or misleading phrasing
- Potential factual errors
- Grammar or formatting issues
- Specific suggestions for improvement

The review results stream in below the questions. Each issue references a specific question number with a concrete suggestion.

Note

AI Review uses the “Review Quiz for Issues” prompt template, which you can customize via the Prompt Editor. You can also add templates for exam review, rubric review, etc.

LaTeX to Canvas Converter

The **LaTeX** → **Canvas** tab solves one of the most frustrating problems in Canvas: getting mathematical equations to display correctly. Canvas does not natively support LaTeX input, but it does render equations via specially encoded image URLs. This tab automates the conversion.

How It Works

The tab provides a side-by-side editor:

- **Left panel:** An HTML editor where you write your page content.
- **Right panel:** A live preview showing how Canvas will render it.

Canvas equations use a special URL format:

```
https://canvas.instructure.com/equation_images/<encoded-latex>
```

where the LaTeX is double-URL-encoded. This tab handles the encoding automatically.

Inserting Equations

1. Type $\LaTeX$ in the **equation bar** (e.g., $\frac{dN}{dM} = kM^{-\alpha}$).
2. A live preview appears showing how the equation will render.
3. Click **Insert Equation** to insert the Canvas-formatted `<img>` tag at the cursor position in the editor.
4. The right panel updates to show the rendered result.

Quick-Insert Symbols

A toolbar provides one-click buttons for common $\LaTeX$ symbols: fractions, square roots, integrals, summations, Greek letters ($\alpha, \beta, \gamma, \lambda$), solar mass ($M_{\odot}$), operators ($\pm, \approx, \leq, \geq, \partial, \nabla$), and more.

Batch Conversion

If you prefer to write content using $\$ \dots \$$ (inline) and $\$ \$ \dots \$ \$$ (display) delimiters, click **Convert All $\$ \dots \$$** to automatically find all $\LaTeX$ delimiters in the editor and replace them with Canvas equation image tags.

Tip

This is the fastest workflow: write your page content using familiar $\LaTeX$ delimiters, then click one button to convert everything to Canvas format.

AI-Assisted $\LaTeX$

Click **AI Assist** and describe the equation you need in plain English (e.g., “Stefan-Boltzmann law” or “integral of x squared from 0 to infinity”). Claude will generate the $\LaTeX$ code and populate the equation bar for you.

Loading and Saving Canvas Pages

- **Load from Canvas Page:** Fetches an existing page’s HTML content into the editor for editing.
- **Save to Canvas Page:** Pushes the editor content directly to a Canvas page.
- **Copy HTML:** Copies the raw HTML to your clipboard for pasting elsewhere.

AI Assistant

The AI Assistant tab has a two-panel layout: a **skills/macros panel** on the left and a **chat interface** on the right.

AI Skills Panel

The left panel shows clickable macro buttons for common AI tasks. Each button corresponds to a prompt template stored in `prompts.json`. The default skills include:

Generate Quiz Question Create quiz from topic/source	Generate Exam Mixed-type comprehensive exam
Generate from Reading Quiz from course material	Review Quiz for Issues Wording, ambiguity, errors
Generate Assignment Full assignment with rubric	Review Assignment Clarity, completeness check

Click a skill → a prompt dialog asks for your input (topic, content, etc.) → the completed prompt is sent to the chat. Responses stream in the chat panel on the right.

The gear icon at the top of the skills panel opens the Prompt Editor, where you can customize any skill's prompt or add new ones.

Chat Interface

The right panel is a free-form chat with Claude. It can:

- Generate quiz questions, exams, and assignment descriptions
- Write rubrics and grading criteria
- Draft course announcements
- Look up student enrollment and course data (via MCP)
- Answer questions about Canvas course content
- Review and critique assessment materials

1. Select a course from the dropdown (gives the AI context about which course to query).
2. Type your request in the text box, click a suggestion chip, or use a skill macro.
3. Press Enter or click Send.
4. The response streams in real-time.

MCP Setup

For the AI to access Canvas data directly, you need a Canvas MCP server configured. Create a `canvas.mcp.json` file in the project directory:

```
{
  "type": "http",
  "url": "https://your-mcp-server/mcp",
  "headers": {
    "X-Canvas-API-Token": "your_token",
    "X-Canvas-API-URL": "https://your-canvas.instructure.com"
  }
}
```

Important

The `canvas.mcp.json` file contains your API token and is excluded from version control by `.gitignore`. Never commit this file.

Prompt Templates

Prompt templates are editable “recipes” that the AI uses for specific tasks. They are stored in `prompts.json` in the project directory and can be edited through the GUI.

Opening the Prompt Editor

Click the **gear icon** () in the application header bar (next to the course selector). This opens a modal with:

- A **list of templates** on the left.
- An **editing panel** on the right showing the selected template’s name, description, and prompt text.

Template Placeholders

Prompt templates use curly-brace placeholders that get filled in at runtime:

Placeholder	Filled with
<code>{topic}</code>	The user’s topic or instructions
<code>{content}</code>	Content being reviewed (quiz text, assignment description)
<code>{source_material}</code>	Concatenated text from selected source materials
<code>{num_questions}</code>	Number of questions to generate
<code>{question_type}</code>	Question type (multiple_choice, essay, etc.)
<code>{points}</code>	Points per question

Example: Custom Review Prompt

Suppose you want a template that specifically checks for Bloom's taxonomy coverage:

Example custom prompt template

```
You are an exam designer for a college-level course. Analyze the
following questions and classify each by Bloom's taxonomy level
(Remember, Understand, Apply, Analyze, Evaluate, Create).
```

```
Questions:
{content}
```

```
For each question:
```

1. State the Bloom's level
2. Explain your classification
3. Suggest how to elevate the question to a higher level

```
Provide a summary table at the end.
```

Default Templates

generate_quiz Generates quiz questions with structured JSON output. Handles MC, T/F, short answer, essay, and mixed types.

generate_from_reading Like **generate_quiz** but emphasizes comprehension over memorization when source material is provided.

generate_assignment Creates a complete assignment with title, HTML instructions, and grading criteria.

generate_exam Creates a comprehensive exam with mixed question types and varied cognitive levels.

review_quiz Identifies ambiguous wording, multiple-correct-answer problems, trick questions, and factual errors.

review_assignment Checks assignment descriptions for clarity, missing information, and grading criteria.

review_exam Assesses coverage, difficulty balance, Bloom's levels, time management, and point distribution.

Adding and Deleting Templates

- Click **+ New Prompt** in the editor header to create a new template. You'll be asked for a key name.
- Fill in the name, description, and prompt text. Use placeholders where appropriate.
- Click **Save** to persist changes to `prompts.json`.

- Click **Delete** to remove a template.

Common Workflows

Weekly Check-In: Are Students On Track?

1. Open the **Students** tab.
2. Check the **Missing Assignments** card — follow up with students who have multiple missing items.
3. Review the **Grade Distribution** — if it's skewed low, consider reviewing the material.
4. Look at **Per-Assignment Averages** — a low average on a specific assignment may indicate unclear instructions.

Preparing Next Week's Content

1. Open the **Calendar** tab to see what's due and when.
2. Switch to **Assignments** to create or edit upcoming assignments. Set due dates and availability windows.
3. Use the **Files** tab to upload any new materials (lecture slides, datasets, readings) by dragging from local to Canvas.
4. Switch to **Quizzes** to create a review quiz. Use **AI Assistant** to generate question ideas.

Creating a Quiz from Scratch

1. Go to the **Quizzes** tab and click **+ New**.
2. Enter a title and set the quiz type (Graded Quiz, Practice Quiz, etc.).
3. Click **Create Quiz**.
4. Click **+ Add Question**. Choose a question type from the grid (e.g., Multiple Choice).
5. Fill in the question text, points, and answers. Click **Add Question**.
6. Repeat for each question — you can mix different types.
7. When ready, click **Publish** to make it available to students.

AI Quiz from Course Readings

1. Go to the **Quizzes** tab and click **AI Generate**.
2. Enter a topic: “Stellar evolution and the Hertzsprung-Russell diagram.”
3. Click **+ Add Sources**, switch to **Local Files**, navigate to your lecture notes PDF, check the box, and click **Add Selected**.
4. Set: 8 questions, Mix, 2 pts each.
5. Click **Generate Questions**. Wait for the AI to produce questions.
6. Review each generated question. Edit wording, fix answers, adjust points, delete weak questions.
7. Click **Create Draft Quiz in Canvas**.
8. The quiz appears in your sidebar. Click **AI Review** to check for issues.
9. Fix any flagged problems, then **Publish**.

AI-Generated Problem Set

1. Go to the **Assignments** tab and click **AI Generate**.
2. Enter: “Problem set on orbital mechanics. Include derivations, numerical calculations, and one Python coding problem.”
3. Add sources: select the relevant Canvas page and a local textbook chapter PDF.
4. Select **Problem Set**, 100 points.
5. Click **Generate Assignment**.
6. Edit the generated HTML description in the split editor. The live preview shows how students will see it.
7. Click **Create Draft Assignment in Canvas**.
8. Open the draft, click **AI Review** for a clarity check, then **Publish**.

Reviewing an Existing Exam

1. Load the quiz/exam in the **Quizzes** tab.
2. Click **AI Review** in the header.
3. The AI streams a detailed review: question-by-question analysis of wording, difficulty, potential issues.
4. For each flagged issue, edit the question text directly in the side-by-side editor.
5. Click **Save** on each modified question.

Uploading Course Materials

1. Go to the **Files** tab.
2. In the left panel, navigate to the file you want to upload. Use the ↑ Up button, Home, or Desktop buttons to navigate.
3. Optionally use the type filter (PDF, Code, etc.) to find specific files.
4. Drag the file onto the target Canvas folder in the right panel.
5. Confirm the upload in the dialog.

Customizing an AI Prompt

1. Click the **gear icon** in the header.
2. Select a template from the left panel (e.g., “Review Quiz for Issues”).
3. Edit the prompt text. For example, add: “Also check that each question maps to a specific learning objective.”
4. Click **Save**.
5. The next time you use AI Review or the skill macro, it will use your updated prompt.

Troubleshooting

Problem	Solution
“API error: 401”	Your Canvas API token is invalid or expired. Generate a new one and update <code>.env</code> .
No courses appear in drop-down	Verify <code>CANVAS_API_URL</code> is correct and includes <code>/api/v1</code> . Ensure your token has teacher enrollment.
AI Assistant: “Claude CLI not found”	Install Claude Code: <code>npm install -g @anthropic-ai/claude-code</code>
AI Assistant: permission errors	Ensure <code>canvas.mcp.json</code> exists in the project directory with correct credentials.
Port 5050 already in use	Another instance is running. Kill it: <code>lsof -ti:5050 xargs kill -9</code>
Student Progress shows no data	The course may not have graded assignments yet, or students haven’t submitted work.

Security Considerations

- The application runs **locally only** on 127.0.0.1:5050. It is not designed for public deployment.
- Your Canvas API token is never sent to any service other than your Canvas instance.
- The `.gitignore` excludes `.env` and `canvas.mcp.json` to prevent accidental token commits.
- The AI Assistant inherits your shell environment. Ensure your Anthropic/Bedrock API keys are set via environment variables, not hardcoded.
- If you share this tool with others, remind them to generate their own API tokens.